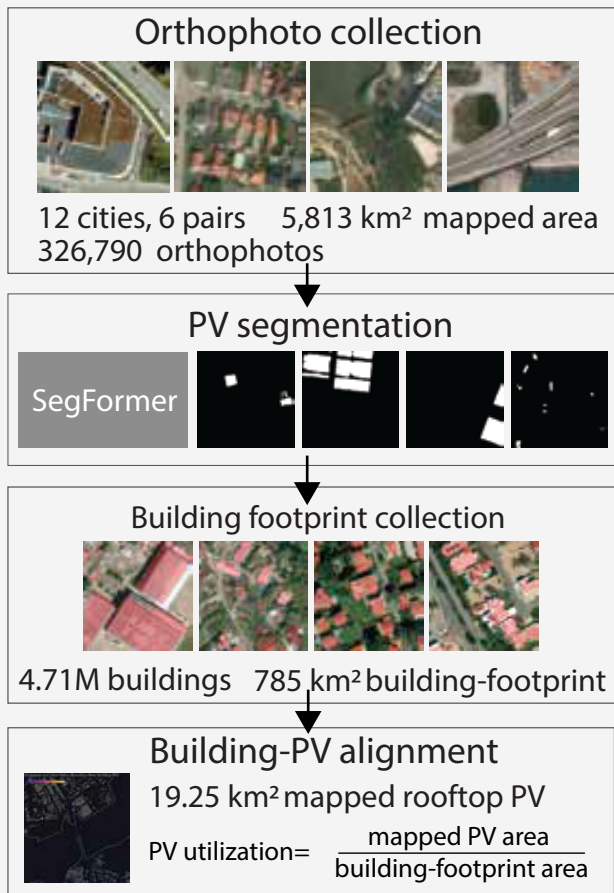
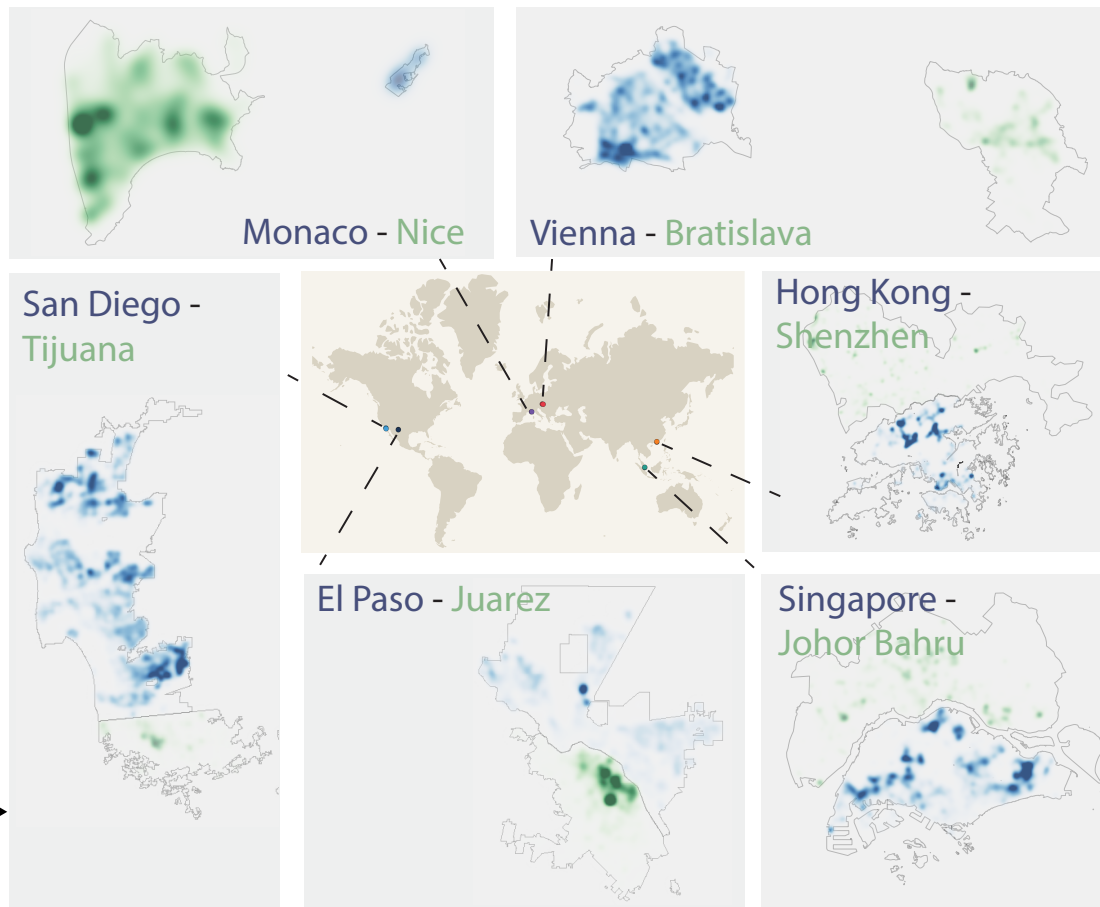


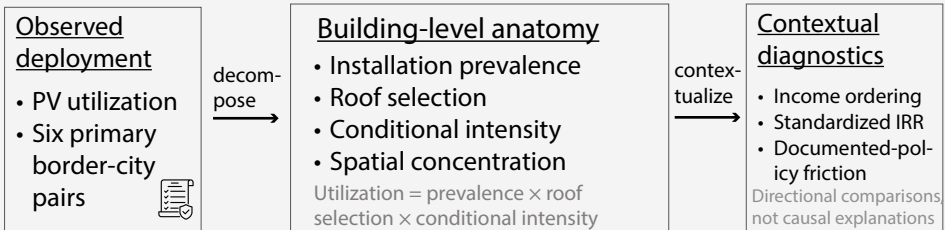
a, Border-city building rooftop PV mapping



b, Mapped rooftop PV density across border-city pairs



c, From citywide gaps to building-level anatomy



d, Two overlapping attributes organize six contrasts

Res. vs Non-res. leadership	All-building-PV vs Income leadership	
	Income-aligned	Income-reversed
Same-side	- Vienna–Bratislava - Singapore–Johor Bahru - San Diego–Tijuana	El Paso–Juarez
Split	<i>No studied pair</i>	- Hong Kong–Shenzhen - Monaco–Nice